GGG-B-191C
May 4, 1992
SUPERSEDING
GGG-B-191B
December 1, 1988

FEDERAL SPECIFICATION

BENDER, TUBE, HAND, (PORTABLE)

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand-operated benders for use in both precision and non-precision cold bending of round metal tubing (pipe, conduit, and heavy steel tubing excluded) used in aircraft hydraulic systems, instrument control lines, fuel and oil lines on engines, refrigeration, plumbing, heating, and other installations (see 6.1).

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. Tube benders shall be of the following types and classes as specified (see 6.2).

Type I - Worm-gear actuated, multiple size
  Class 1 - Small radius bending
  Class 2 - Large radius bending

Type III - Lever actuated
  Class 1 - Multiple size
  Class 2 - Single size

Type IV - External coil spring, hand-actuated

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatment for Organic Coatings
TT-E-485	- Enamel, Semigloss, Rust-Inhibiting
PPP-P-40	- Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-376	- Preferred Metric Units for General Use by the Federal Government
FED-STD-595	- Colors Used in Government Procurement

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Specifications:

MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

A 251 - Standard Specification for General Requirement Wrought Seamless Copper and Copper Alloy Tube
A 269 - Seamless and Welded Austenitic Stainless Steel Tubing for General Service
B 571 - Test Methods for Adhesion of Metallic Coatings

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of tube benders otherwise conforming to this specification.

3.2 Material. Material used in construction of tube benders shall be of such composition and heat-treatment to preclude appreciable deformation of any part when subjected to routine, sustained usage. Tube benders shall be tested for the applicable characteristics specified in 4.3.

3.3 Construction. Tube benders shall be so constructed that no assembled part will become detached while in use. All moving parts shall operate without binding and shall be constructed so that excessive looseness and play adversely affecting operation will not develop due to routine, sustained usage.

3.4 Design. Tube benders shall be of the latest design. All parts subject to wear and normally replaceable shall be readily accessible for adjustment or repair (see 6.2.1).

3.5 Threads. All threaded parts shall be of a nationally recognized commercial standard thread having standard size, shape, and tolerances in order to insure correct mating of parts of interchangeability and replacement.

3.6 Lubrication. All bearings and surfaces requiring periodic lubrication shall be provided with accessible means for lubricating.

3.7 Surface finish. All castings, forgings, or welded parts shall be thoroughly cleaned and shall be free of sand, dirt, fins, sprues, scale, flux, or other harmful or extraneous foreign materials. All external surfaces and edges shall be free of burrs, fins, and sharp edges.

3.8 Corrosion-resisting coatings. Type IV benders and material components of all other types of benders which are subject to corrosion, shall be coated with either zinc, hot phosphate, or oxide in accordance with accepted industry practice. Chromium finish may be used providing that it meets the surface adhesion requirements specified in 4.3.5.

3.9 Interchangeability of parts. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

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3.10 Operating instructions. All tube benders, except type IV, shall be furnished with two sets of printed instructions covering operation and maintenance. When carrying cases are furnished, the instructions shall be enclosed within; otherwise, they shall be attached to the bender.

3.11 Marking. All tube benders, except type IV, shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item. For military purchases, marking shall be in accordance with MIL-STD-130, as practicable.

3.12 Type I, worm-gear actuated, multiple size. Type I tube benders shall be designed for making precision bends on cold-drawn, corrosion-resisting steel tubing conforming to AISI type 304 of ASTM A 269 without the aid of mandrels.

3.12.1 Type I operational and functional design. The operating mechanism of type I benders shall have mechanical advantage to provide a smooth, continuous operation in producing bends of approximately 180 degrees on tubing of the diameter shown in the applicable table without subjecting an operator to undue strain or fatigue. The design shall be of such that, after making a bend of approximately 180 degrees, the tubing can be easily removed from the bender. Means shall be provided in accordance with industry practice for clamping the bender in a standard machinist's bench vise having a jaw opening of 4 inches or more.

3.12.2 Overload pin or safety device. Each tube bender shall be provided with an overload pin or safety device in the event failure occurs due to overloading. An attempt to bend heavy-walled tubing beyond the capacity of the bender would only result in shearing the overload pin which could be readily replaced and prevent serious damage to the bender.

3.12.3 Type I angle of bends. The bender shall bend the specified tubing to any angle from 0 degrees to approximately 180 degrees without having to reset or adjust the tubing. Visible means shall be provided on the bender for measuring angles of bend in 15-degree increments up to approximately 180 degrees. Benders shall be capable of being positioned at any point on the tubing and also making various styles of bends such as offset, return, and right angle.

3.12.4 Type I quality of bends. The bender shall accomplish the desired bends without marring the bent section with diemarks, kinks, wrinkles, or surface imperfections. Unless otherwise specified, the maximum reduction in cross-sectional diameter at any point on the bend shall not exceed 10 percent of the original cross-sectional diameter of the tubing when bends are made within the radius range shown in the applicable table.

3.12.5 Weight and dimensions. The combined weight of the tube bender, accessories, and carrying case shall not exceed 40 pounds for class 1 bender and 45 pounds for class 2 bender. The dimensions of the carrying case shall not exceed 16-1/4 by 9-1/2 by 8 inches, including hasps, hinges, and handles.

3.12.6 Type I carrying case. Each tube bender and its accessories shall be furnished in a metal carrying case with compartments or sections for securing loose parts within the case. The case shall have a hinged lid with at least one draw-bolt or other type of catch and a full-grip handle securely attached to the lid. Both the interior and exterior surfaces of the case shall have at least one protective coat of paint. The case shall be finished as specified in 3.15.

3.12.7 Type I identification marking. Parts used in forming the bends such as sheaves, form blocks, slide blocks, or wiping dies shall be marked in a plain and permanent manner with the outside diameter of the tubing for which the part is used.

3.12.8 Class 1, small radius bending. Class 1 tube bender shall be furnished with all necessary parts for bending all sizes of the tubing specified in table I. The bender shall be similar to figure 1.

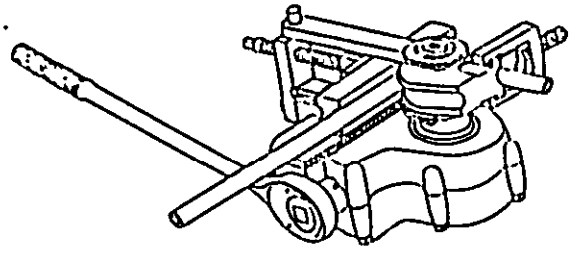


Figure 1. Type I, class 1, tube bender, worm-gear actuated, multiple size, small radius bending.

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TABLE I. Type I, class 1, tube bender, worm-gear actuated, multiple size, small radius bending

Outside diameter of tubing	Range of wall thickness of the tubing	Radius of bend measured to centerline of the tubing (+ 1/16, - 1/8)
Inch	Inch	Inches
1/4	0.016 - 0.049	9/16
5/16	0.020 - 0.065	11/16
3/8	0.028 - 0.083	15/16
1/2	0.035 - 0.083	1-1/4
5/8	0.042 - 0.083	1-1/2
3/4	0.049 - 0.083	1-3/4

3.12.9 Class 2, large radius bending. Class 2 tube benders shall be furnished with all necessary parts for bending all sizes of the tubing specified in table II. The bender shall be similar to figure 2.

TABLE II. Type I, class 2, tube bender, worm-gear actuated, multiple size, large radius bending

Outside diameter of tubing	Range of wall thickness of the tubing	Radius of bend measured to centerline of the tubing (+ 1/16, - 1/8)
Inch	Inch	Inches
1/4	0.016 - 0.049	3/4
5/16	0.020 - 0.065	1
3/8	0.028 - 0.083	1-1/4
1/2	0.035 - 0.083	2
5/8	0.042 - 0.083	2-1/2
3/4	0.049 - 0.083	3

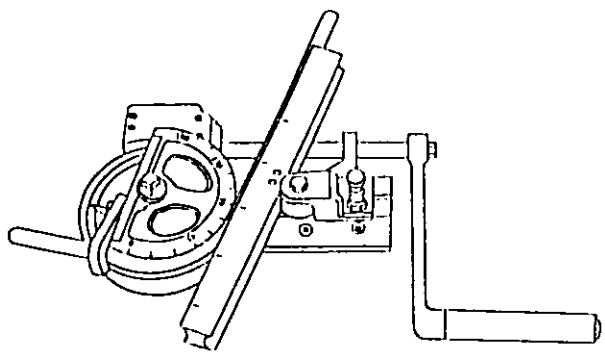


Figure 2. Type I, class 2, tube bender, worm-gear actuated, multiple size, large radius bending.

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3.13 Type III, lever actuated.

3.13.1 Class 1, multiple size. Class 1 tube benders shall be designed for making accurate bends on soft-copper, aluminum, steel, or other soft-metal tubing. When specified (see 6.2), the bender shall also be capable of bending hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy steel tubing. The maximum wall thicknesses for 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, and 7/8 inch sizes of tubing which class 1 benders will bend without deformation of the bender and without undue effort by the operator are as follows: 0.049, 0.049, 0.055, 0.060, 0.060, 0.065, and 0.065 inch, respectively. Type III, class 1 benders shall be furnished with all necessary parts for bending the applicable sizes of tubing listed for that particular bender size number in table III and shall be similar to figure 3.

TABLE III. Type III, class 1 tube bender, lever actuated, multiple size

Bender size number	Tubing capacity outside diameter of the tubing	Minimum wall thickness without exceeding 10% reduction of cross-sectional diameter	Radius of bend measured to centerline of the tubing		Weight of bender complete (includes carrying case when furnished) (see 3.13.1.4)
			Min.	Max.	Max.
	Inch	Inch	Inches	Inches	Pounds
1	1/2	0.035	1-1/2	2-1/8	16-1/2
	5/8	0.035	2-1/8	2-5/8	
2	3/8	0.035	1-1/2	1-3/4	27
	1/2	0.035	1-1/2	1-3/4	
	5/8	0.035	2-1/8	2-3/8	
	3/4	0.035	2-7/8	3-1/8	
3	1/4	0.028	9/16	7/8	29-1/2
	5/16	0.028	11/16	1-1/8	
	3/8	0.035	1-1/8	1-3/4	
	1/2	0.035	1-1/2	1-3/4	
	5/8	0.035	2-1/8	2-3/8	
	3/4	0.035	2-7/8	3-1/8	
4	1/4	0.028	9/16	7/8	34
	5/16	0.028	11/16	1-1/8	
	3/8	0.035	1-1/8	1-3/4	
	1/2	0.035	1-1/2	1-3/4	
	5/8	0.035	2-1/8	2-3/8	
	3/4	0.035	2-7/8	3-1/8	
	7/8	0.042	2-7/8	3-1/8	
5	5/8	0.035	2-1/8	2-5/8	20
	7/8	0.042	2-7/8	3-1/8	
6	3/4	0.035	2-7/8	3-1/8	20
	7/8	0.042	2-7/8	3-1/8	
7	5/8	0.035	2-1/8	2-5/8	20
	3/4	0.035	2-7/8	3-1/8	

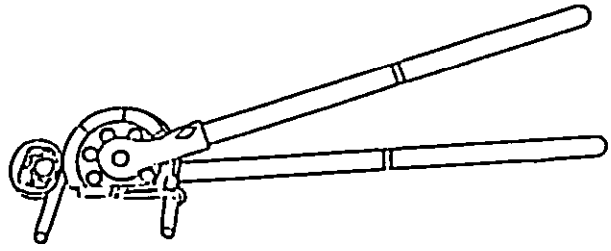


Figure 3. Type III, class 1 tube bender, lever actuated, multiple size.

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3.13.1.1 Type III, class 1, operational and functional design. The benders shall have leverage to maintain a smooth performance throughout the bending operation. The benders shall be of the open-side type and capable of being positioned on the tubing at any point where a bend is required. The benders shall be so designed that the tubing can be easily freed after bending. The benders shall also be designed for medium heavy-duty performance with provisions made for clamping in a bench vise.

3.13.1.2 Type III, class 1, angle of bends. The bender shall be capable of bending tubing of the diameter of 3/8 to 3/4 inch to any angle from 0 degrees to 180 degrees without having to reset or adjust the tubing. Visible means shall be provided on the bender for measuring angles of bend in 45-degree increments up to 180 degrees.

3.13.1.3 Type III, class 1, quality of bends. The bender shall be capable of accomplishing the desired bends without kinks, wrinkles, or scratches which would likely contribute to failure of the bent section in service. The maximum reduction in cross-sectional diameter at any point on the bend shall not exceed 10 percent of the original diameter of the tubing when bends are made on soft tubing having a wall thickness equal to or greater than that specified in table III.

3.13.1.4 Type III, class 1, carrying case. A carrying case shall be furnished with each size number 2, 3, and 4 tube bender. The case shall conform to 3.12.6, and the maximum dimensions shall be 21-1/2 by 8-1/2 by 8-1/2 inches. The case shall be finished as specified in 3.15.

3.13.1.5 Type III, class 1, identification marking. Forming wheels, sheaves, or blocks shall be marked in a plain and permanent manner with the outside diameter of the tubing for which the part is designed.

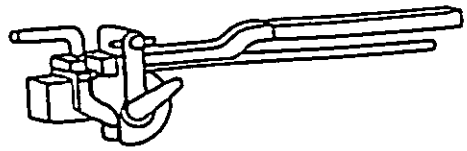
3.13.2 Class 2, single size. Class 2 tube benders shall be designed for making accurate bends on soft-copper, aluminum, or steel tubing, except that the 1 inch size is not recommended for steel tubing. When specified (see 6.2), benders in sizes 1/8 inch through 7/8 inch shall be capable of bending hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy steel tubing. The maximum wall thicknesses for 1/8, 3/16, 1/4, 5/16, 3/8, 7/16, 1/2, 5/8, 3/4, 7/8, and 1 inch sizes of tubing which class 2 benders will bend without deformation of the bender and without undue effort by the operator are as follows: 0.042, 0.049, 0.049, 0.049, 0.055, 0.055, 0.060, 0.060, 0.065, 0.065, and 0.083 inch respectively. Type III, class 2 tube benders shall conform to the requirements shown in table IV, and shall be similar to figure 4 (see figure 4a, 4b, 4c, or 4d). For the 4d bender, a swivel handle is incorporated to help facilitate 180 degree bends.

TABLE IV. Type III, class 2 tube bender, lever actuated, single size

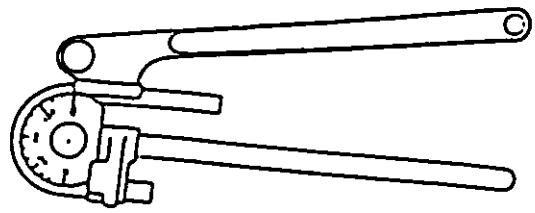
Bender size and outside diameter of tubing	Minimum wall thickness without exceeding 10% reduction of cross-sectional diameter	Radius of bend measured to centerline of the tubing		Weight (maximum)
		Min.	Max.	
Inch	Inch	Inches	Inches	Pounds
1/8	0.028	3/8	7/16	5/8
3/16	0.028	7/16	5/8	1-3/4
1/4	0.032	9/16	5/8	4
5/16	0.032	11/16	15/16	4-3/4
3/8	0.035	7/8	1	5-1/2
7/16	0.042	1	1-1/2	7
1/2	0.042	1-1/4	1-1/2	7
5/8	0.042	1-1/2	2-1/4	10
3/4	0.049	1-3/4	3	16
7/8	0.049	2	3	19
1	0.065	2-1/2	3-1/2	32

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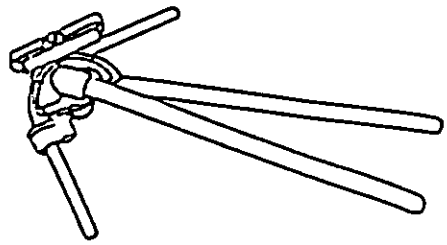
(a)



(b)



(c)



(d)

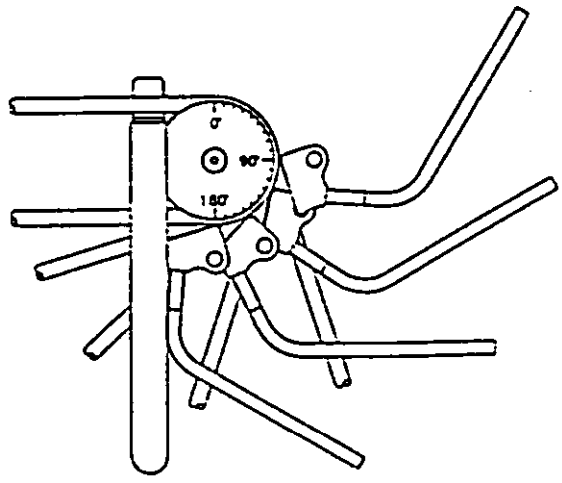


Figure 4. Type III, class 2 tube bender, lever actuated, single size.

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3.13.2.1 Type III, class 2, operational and functional design. The benders shall have leverage to maintain a smooth performance throughout the bending operation. They shall be capable of producing bends of 180 degrees and of being positioned on the tubing at any point where a bend is required.

3.13.2.2 Type III, class 2, angle of bends. The benders shall be capable of bending tubing to any angle from 0 degrees to 180 degrees without having to reset or adjust the tubing. Visible means shall be provided on the bender for measuring angles of bends in accordance with the manufacturer's standard practice.

3.13.2.3 Type III, class 2, quality of bends. The bender shall be capable of accomplishing bends in conformance with 3.13.1.3 except that the wall thickness of tubing shall be equal to or greater than that shown in table IV.

3.13.2.4 Type III, class 2, identification marking. Each tube bender shall be marked in plain and permanent manner with the outside diameter of the tubing for which the bender is designed.

3.14 Type IV, external coil spring, hand actuated. The type IV tube benders shall be designed for use in making non-precision bends on soft-copper and aluminum tubing. Type IV tube benders shall conform to table V and shall be similar to figure 5.

TABLE V. Type IV tube bender, external coil spring, hand actuated

Bender sizes and outside diameter of tubing	Overall length		Weight	
	Min.	Max.	Min.	Max.
Inch	Inches	Inches	Ounces	Ounces
1/4	10	11	1-1/2	3-1/2
5/16	10	11	2-1/2	3-1/2
3/8	10	11	3-1/2	4-1/2
7/16	12	13	6	7
1/2	12	13	6	7-1/2
5/8	12	13	7-1/2	9

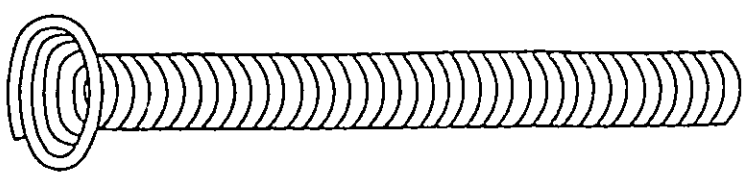


Figure 5. Type IV tube bender, external coil spring, hand actuated.

3.14.1 Type IV material. The benders shall be hard-drawn spring-wire, oil-tempered, or piano wire and shall be strain relieved after coiling. The diameter of the wire shall be that which is regularly used for standard tube benders of this type.

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3.14.2 Type IV operational and functional design. At least one end of each bend shall be flared or other means shall be provided in order that the bender may be easily removed from the tubing after a bend is made. The coil spring shall have clearance to slide over the tubing easily. The minimum and maximum wall thicknesses for 1/4, 5/16, 3/8, 7/16, 1/2, and 5/8 inch sizes of tubing which the coil spring bender will support sufficiently to prevent collapse of the tubing when a bend is made are as follows: 0.030 - 0.035, 0.032 - 0.035, 0.032 - 0.035, 0.032 - 0.035, 0.032 - 0.035, and 0.035 - 0.040 inch, respectively. The design of the bender shall be such that soft tubing will not be scratched.

3.14.3 Sets. Type IV benders shall be furnished in individual sizes or in sets of four or six sizes, as specified (see 6.2). Sets of four shall consist of sizes 1/4, 3/8, 1/2, and 5/8 inch. Sets of six shall consist of all sizes shown in table V.

3.14.4 Chest. When specified (see 6.2), sets of four or six benders shall be furnished in a commercially constructed wood or steel chest having a hinged lid and a fastener. The chest shall be finished as specified in 3.15.

3.15 Finish.

3.15.1 Protective coating. Unless otherwise specified (see 6.2), the finish and color of paint used on the interior and exterior surfaces of the carrying cases or chests shall be in accordance with manufacturer's commercial practice. When specified (see 6.2), the cases or chests shall be furnished with either olive drab or gray enamel as specified in 3.15.2 or 3.15.3.

3.15.2 Olive drab. When olive drab finish is specified, the interior and exterior surfaces of the cases or chests shall be prepared for painting in accordance with any method of TT-C-490 and then shall be coated with enamel conforming to TT-E-485 and shall match color No. 24084 of FED-STD-595.

3.15.3 Gray enamel. When gray finish is specified, the interior and exterior of the cases or chests shall be coated with light gray equipment enamel in accordance with formula 111, type III, class 2 of MIL-E-15090.

3.16 Workmanship. Edges and corners of tube benders shall be curved in conformance with the best commercial practice. Benders shall have no burrs, fins, sharp projections, excessive surface roughness, or any imperfections which may impair their serviceability.

3.17 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

3.18 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Federal Specification are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Visual examination. The tube benders shall be examined for the defects listed in table VI. The lot size shall be expressed in terms of tube benders. The inspection level shall be I, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

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TABLE VI. Classification of defects

Examine	Defects
Construction and Workmanship	Type not as specified
	Class not as specified (types I and III)
	Size not as specified (type III); set not as specified (type IV)
	Construction and design nonconforming; not actuated as specified, worm and worm-gear fail to mesh properly; bender not convenient to operate (type I)
	Bender incomplete, component parts missing, damaged, improperly formed or positioned; parts not suitably secured to preclude becoming detached or development of excessive looseness or play during hard usage (types I and III)
	Moving parts fail to operate without binding or excessive looseness and play (types I and III)
	Worm or worm-gear teeth damaged, cracked, broken, chipped, or improperly finished (type I)
	Parts subject to wear and normal replacement not readily accessible for adjustment or repair (types I and III)
	Accessible means for lubricating not provided for all bearings and surfaces requiring lubrication (types I and III)
	Tubing cannot be easily removed from bender after making the required maximum bend (types I and III)
	Adequate means not provided for clamping the bender in a standard machinists' bench vise having the designated minimum jaw opening (type I)
	Bender not provided with an overload pin or safety device for adequate protection against overloading; overload pin, when sheared in bending tubing beyond the bender capacity, not readily replaceable (type I)
	Bender not capable of bending the applicable tubing to any angle within the stipulated range without having to reset or adjust the tubing
	Visible means not provided on bender for measuring angles of bend in the designated increments up to the specified maximum angle (types I and III)
	Bender not capable of being positioned at any point on the tubing or making the various stipulated styles of bends (type I)
	All necessary parts for bending each of the specified sizes of tubing not furnished (type I and type II, class 1)
	Carrying case not furnished; not suitably made of metal with rigidity to withstand the required handling; lid not hinged with a catch and a full grip handle not provided or not securely attached; means not provided for securing loose parts within the case, when required (types I and III)
	Threads nonconforming, stripped, crossed, chipped, or torn (types I and III)
	Replacement parts not dimensionally and functionally interchangeable without requiring modification (types I and III)
	Construction and design nonconforming; not actuated as specified; bender not convenient to operate (type III)
	Lever handles not removable (type III, class I)
	Bender not open-side type, cannot be positioned on tubing at any point where a bend is required (type III)
	Provision not made for clamping in a bench vise, when applicable, (type III)
	Printed instructions covering operation and maintenance not enclosed in carrying case or attached to bender, as applicable
	End not flared, edges and corners not curved (type IV)
	Inside diameter of the coil spring fails to have the proper clearance for the applicable diameter of tubing (type IV)
	Chest not furnished for set when specified; fastener not provided for lid or lid not hinged (type IV)

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TABLE VI. Classification of defects (cont'd)

Examine	Defects
Material	Not suitable for intended use; not sound and uniform; not clean; not free from sand, dirt, fins, sprues, scale, flux, or other injurious defects (types I and III) Nonconforming, not spring tempered, not smooth and uniform, not free from cracks, splits, diemarks, burrs, fins, sharp projections, or excessive surface roughness (type IV) Wire diameter inadequate, not uniformly coil wound, deformed or not shaped true and round (type IV)
Finish	Component parts subject to corrosion not coated as required, coating not smooth, uniform, and adherent (types I and III) Interior and exterior surfaces of carrying case not painted as required, when applicable (types I and III) Working surfaces of sheaves, form blocks, slide blocks, or wiping dies not smoothly finished or suitably formed, not free from galling, chatter, toolmarks, gouges, nicks, or dents (type I) External surfaces and edges not free from burrs, fins, and sharp edges (type I) Working surfaces of forming wheel(s), sheave(s), or block(s) not smoothly finished or suitably formed; not free from galling, chatter, toolmarks, gouges, nicks, or dents (type III)
Marking	Parts used in forming bends not marked in a plain and permanent manner with the applicable outside diameter of the tubing for which the part is used (type I and type III, class 1) Bender not marked in a plain and permanent manner with the applicable outside diameter of the tubing for which the bender is designed (type III) Manufacturer's name or trademark missing, incorrect, illegible, or not permanent (type I and III)

4.2.2 Dimensional examination. The end items shall be examined for conformance to dimensions specified herein. The lot size shall be expressed in terms of tube benders. The sample unit shall be one tube bender. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.3 Testing. Each sample unit shall be tested in accordance with 4.3 through 4.3.5, as applicable. The sample unit shall be one completely fabricated bender. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.3 Testing methods. Filling of the tubing with sand, sulfur, or other material will not be permitted in performing the tests specified in 4.3.1 through 4.3.5.

4.3.1 Test for type I tube benders. Cold-drawn corrosion-resisting steel tubing conforming to AISI type 304 of ASTM A 269 shall be used for this test. At least two bends of 180 degrees shall be made on each of the six outside diameters of tubing specified for the bender, one each of the minimum and maximum wall thicknesses as specified in the applicable table, making a total of 12 bends to be made with each bender. The benders shall show no evidence of permanent deformation, binding of mechanism, excessive pressure required for operation, excessive looseness adversely affecting operation, any assembled part becoming detached, difficulty of adjusting or fitting replaceable parts, difficulty in removing the bent tubing, failure to conform to quality of bends specified in 3.12.4, or failure of bends to be within the range or radii specified in the applicable table.

4.3.2 Test for type III, class 1, tube benders. Commercial copper tubing shall be used for this test. At least two bends of 180 degrees shall be made on each of the outside diameters of tubing specified for the applicable size bender, one each at both a minimum wall thickness of 0.035 inch, and a maximum wall thickness of 0.065 inch (see 3.13.1 and table III). When the contract or order specifies that the bender will be required to bend hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy steel tubing (3.13.1), at least one additional bend shall be made on each of the outside diameters specified for the applicable size bender, using hard-drawn copper with a wall thickness at least 20 percent greater than the minimum specified in table III. The quality of bends shall conform to 3.13.1.3 and performance shall conform to 4.3.1. Radius of bends shall conform to table III.

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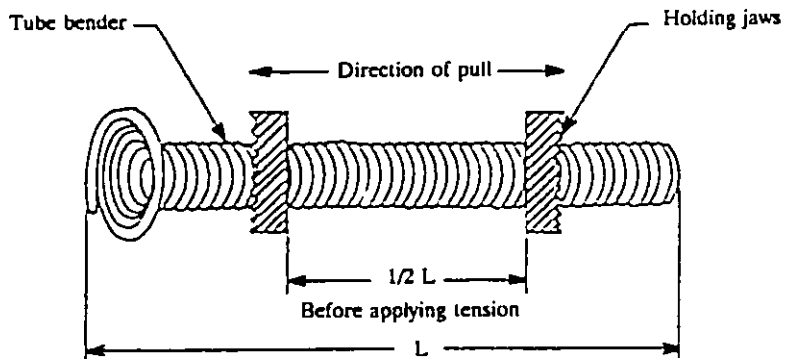
4.3.3 Test for type III, class 2 tube benders. At least two bends of 180 degrees shall be made with each of the benders, using soft commercial copper tubing, one each of the minimum and maximum wall thicknesses specified in table VII. The quality of the bends shall conform to 3.13.2.3 and performance shall conform to 4.3.1. Radius of bends shall conform to table IV. When the contract or order specifies that the bender will be required to bend hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy duty steel tubing (see 3.13.2), at least one additional bend shall be made on each size tube specified for the applicable size bender (see table IV) using hard-drawn copper with a thickness at least 20 percent greater than the minimum specified in table IV. Tolerances for wall thickness shall be in accordance with ASTM A 251.

TABLE VII. Tube testing requirements for type III, class 2 tube bender, lever actuated, single size

Outside diameter of tubing (inches)	Wall thickness (minimum)	Wall thickness (maximum)
1/8	0.032	•
3/16	0.032	•
1/4	0.032	0.048
5/16	0.032	0.048
3/8	0.032	0.048
7/16	0.048	0.062
1/2	0.048	0.062
5/8	0.048	0.062
3/4	0.048	0.062
7/8	0.048	0.062
1	0.062	•

• Only one minimum wall size available for testing.

4.3.4 Tests for type IV tube benders. The tube benders shall not break when held firmly and stretched twice the distance between the holding jaws for 1/4, 5/16, and 3/8 inch sizes, and two and one-half times the distance between the holding jaws for the 7/16, 1/2, and 5/8 inch sizes, in a testing device similar to the one shown in figure 6. After releasing, the overall length shall not exceed the original by more than 3/16 inch. Each bender shall then be used to make a bend of 180 degrees around a mandrel of the size shown on figure 7 for the applicable size soft-copper tubing. The wall thickness of the copper tubing shall be within the range specified in 3.14.2. Removal of the bender shall not cause the bender to break or exceed the original length more than 1/8 inch in addition to the 3/16 inch allowed for the pull test.



L = Length of tube bender

Figure 6. Pull test diagram.

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Mandrel Diameter = D

Maximum Bend Radius = R

Tubing O.D.	R	D
1/4	3/4	1-1/2
5/16	15/16	1-7/8
3/8	1-1/8	2-1/4
7/16	1-5/16	2-5/8
1/2	1-1/2	3
5/8	1-7/8	3-3/4

Dimensions in inches

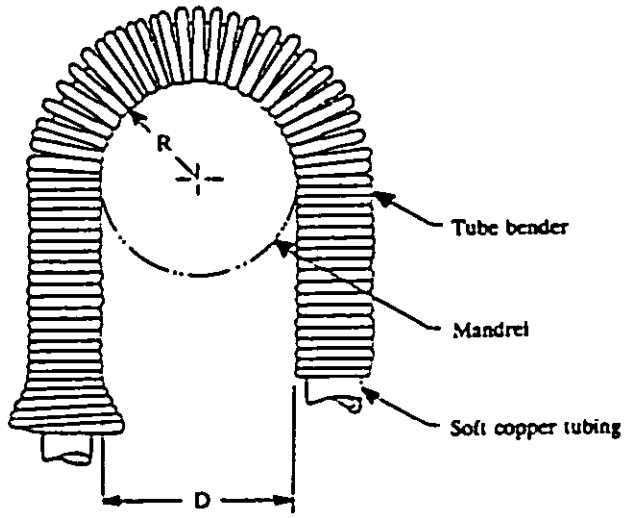


Figure 7. Bend test diagram.

4.3.5 Chromium finish. Chromium finish specified in 3.8 shall be tested in accordance with ASTM B 571, paragraphs 9.1 through 9.4, and table I, Temperature Test Guide.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VIII. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE VIII. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use.

6.1.1 Type I. Type I tube benders are primarily used for making precision bends on cold-drawn corrosion-resisting steel tubing used in aircraft hydraulic systems but can be used for bending many other kinds of metal tubing. Tubing with rough outside surfaces should not be used, as the sheave wheels or blocks may become marred. The radius of a bend on cold-drawn steel tubing should not be less than the minimum specified in table I, as the out surface of the tubing may be stretched beyond the allowable elongation of the material.

6.1.2 Type III. Type III tube benders are used for making accurate bends on any soft tubing. When specified (see 3.13.1 and 3.13.2), type III benders are also available for bending hard-drawn copper, annealed corrosion-resistant steel, and medium-heavy steel tubing. Such benders, however, require considerable effort or extra long handles when bending such tubing in the larger sizes.

6.1.3 Type IV. Type IV tube benders are used for bending soft copper and aluminum tubing to various shapes and as an aid in preventing the tubing from collapsing in making non-precision bends.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class, if applicable, of item required (see 1.2).
c. If the bender will be required to bend hard-drawn copper, annealed corrosion-resisting steel, and medium-heavy steel tubing in addition to soft-metal tubing (see 3.13.1 and 3.13.2).
d. Bender size required (see tables III, IV, or V). For sets of type IV benders, see 3.14.3 (tables I and II not applicable).
e. If a set of type IV benders is required, specify whether the set shall consist of four sizes or six sizes (see 3.14.3).
f. If a chest is required for sets of four or six type IV benders (see 3.14.4).
g. Finish and color for carrying cases or chests, where applicable, if other than specified (see 3.15.1).
h. Level of preservation, packaging, and packing required (see 5.1).

6.2.1 Repair parts. When ordering repair parts, specify the following:

a. The manufacturer's name, or trademark of the tool for which the parts are intended.
b. Type, class, and size of tool, if applicable.
c. The name or description of the part, and size, if applicable.
d. Quantity of each part required.

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Navy - SH
Air Force - 99

Review Activities

Navy - MC
Air Force - 84

User Activity

Navy - YD

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

GSA - FSS

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